Lucas Zheng

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Education

Brown University — Providence, RI
B.S. in Computer Science & B.A. in Applied Mathematics; GPA: 4.00/4.00 — *Aug. 2023 – May 2027*

- Relevant Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Sequential Decision Making

Experience

Applied Machine Learning Intern — Jun. 2026 – Present
Shopify — *New York, NY*

- Built an end-to-end, locale-aware evaluation system gating AI-generated sales-outreach emails, spanning labeled eval-set generation, LLM-judge scoring, and a feedback loop turning reviewer annotations into prompt revisions
- Extended the gate to 6 non-US locales covering 415K outreach emails per year, improving judge agreement with ground truth by 5% through model and context-rotation upgrades

Research Assistant — Sep. 2025 – Present
Intelligent Robot Lab, Brown University — *Providence, RI*

- Building a hierarchical reinforcement learning system under Prof. George Konidaris with concurrent training of option-selection and option-policy agents (DQN, PPO), evaluating learned and fixed options on Montezuma's Revenge and MiniGrid
- Researching exploration strategies including count-based intrinsic motivation methods to address sparse-reward, long-horizon challenges in Montezuma's Revenge where standard exploration fails

Software Engineering Intern — Jul. 2025 – Aug. 2025
Supio — *Seattle, WA*

- Reduced deposition transcript turnaround time from 10 days to minutes by engineering an end-to-end pipeline with Whisper, pyannote speaker diarization, and an OpenAI API correction workflow
- Integrated transcripts into hybrid RAG search system combining pgvector semantic similarity with BM25 ranking algorithms, enabling lawyers to query hours of audio content with sub-second response times

Research Intern — Jul. 2024 – Aug. 2024
Machine Creativity Lab, Hong Kong University of Science and Technology — *Hong Kong*

- Advanced lab's generative music AI's capabilities by training an audio emotion classifier with PyTorch and boosting classification accuracy by 6% through fine-tuning
- Engineered a feature extraction pipeline processing 10,000+ audio samples with automated labeling, later integrated into the lab's production music AI model

Projects

Latent Diffusion World Models & Policies | *Python, PyTorch, Diffusion Models, SLURM* — 2026 – Present

- Investigating when generative (diffusion) models outperform deterministic baselines in latent-space planning, testing the hypothesis that the advantage appears only when the target distribution is multimodal
- Built a latent diffusion policy and dynamics stack with a CEM planner over a frozen vision encoder, beating MSE-regression baselines by +9% closed-loop reward on multimodal action data while tying on unimodal data

Physics-Informed Neural Network for Seismic Localization | *Python, PyTorch, Gymnasium* — 2025

- Developed a Physics-Informed Neural Network (PINN) with Fourier Feature Encoding to solve the Eikonal equation for seismic localization, with a PPO agent optimizing adaptive collocation sampling

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL
Tools & Frameworks: PyTorch, pandas, Gymnasium, scikit-learn, PostgreSQL, MCP, Git, REST APIs

Miscellaneous

Hackathons: 2nd Place winner at 2026 Precision Neuroscience BCI Hackathon
Languages: Mandarin (fluent), Arabic (working proficiency – U.S. State Department NSLI-Y Scholar)
Interests: Squash, Golf, Poker, Chess